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Cart stabilization devices, systems, and methods

Abstract

Devices, assemblies, systems, and methods are disclosed for stabilizing a cart. An example cart is a surgical cart having a robotic arm thereon. A stabilizer system may be part of or used with the cart to stabilize the cart at a location. The stabilizer system may include a stabilizer and an actuator. The stabilizer may have a foot and a biaser configured to bias the foot to a retracted position and contribute to an amount of force applied to a floor supporting the cart when the foot is in a deployed position. The actuator acts on the stabilizer to overcome a bias force biasing the stabilizer to the retracted position and cause feet of the stabilizer to contact the floor. Once the feet of the stabilizer contact the floor, a spring of the biaser causes the foot to apply a predetermined force amount to the floor.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1200022	12/1915	Price	267/168	B60G 11/36
2922494	12/1959	Clark, Jr.	188/5	B60B 33/0089
4833972	12/1988	Bohusch	91/465	A61G 13/10
6336524	12/2001	Van Loon	188/19	B62B 5/0485
12049248	12/2023	Love	N/A	A61B 50/13
2011/0303497	12/2010	Gaffney	188/106P	B60T 7/042
2015/0096815	12/2014	Ottenweller	180/19.1	A61G 7/0528
2018/0304113	12/2017	Goldish et al.	N/A	N/A
2022/0153364	12/2021	Nahum	N/A	B62B 5/049

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
110722927	12/2019	CN	N/A
20180007882	12/2017	KR	N/A
2016145044	12/2015	WO	N/A
2018154750	12/2017	WO	N/A

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of U.S. patent application Ser. No. 17/461,342 (published as U.S. 2023-0062720), filed Aug. 30, 2021, the entirety of which is hereby incorporated by reference.

BACKGROUND

(1) A wide variety of surgical devices and systems have been developed for medical use, for example, surgical use. Some of these devices and systems include carts for use in surgical procedures, among other devices and systems. These devices and systems are manufactured by any one of a variety of different manufacturing methods and may be used according to any one of a

variety of methods. Of the known surgical devices, systems, and methods, each has certain advantages and disadvantages. There is an ongoing need to provide alternative surgical devices and systems, as well as alternative methods for manufacturing and using the surgical devices and systems.

BRIEF SUMMARY

(2) This disclosure provides design, material, manufacturing method, and use alternatives for devices, systems, assemblies used in medical procedures. An example assembly includes a surgical cart having locking caster wheels configured to contact a floor, a stabilizer, and an actuator configured to drive the stabilizer toward a deployed position. The stabilizer may include a foot configured to resist movement of the surgical cart when the foot is in a deployed position contacting the floor, and a biaser configured to bias the foot in a retracted position and contribute to a predetermined amount of force applied to the floor when the foot is in the deployed position. Actuation of the actuator causes the stabilizer to adjust to the deployed position and apply the predetermined amount of force to the floor.

(3) Alternatively or additionally to any of the embodiments in this section, the biaser comprises a first spring biasing the stabilizer to the retracted position, and a second spring biasing the stabilizer to the deployed position.

(4) Alternatively or additionally to any of the embodiments in this section, the biaser comprises a housing, wherein the first spring extends along an exterior of the housing and the second spring extends along an interior of the housing.

(5) Alternatively or additionally to any of the embodiments in this section, the surgical cart comprises a surgical cart base, and the biaser comprises a biaser housing, and a spring configured to act on the surgical cart base and the biaser housing to bias the stabilizer in the retracted position.

(6) Alternatively or additionally to any of the embodiments in this section, the stabilizer comprises a rod, a plate fixed relative to the rod, and a foot connected to the rod and configured to contact the floor when the stabilizer is in the deployed position, wherein the biaser comprises a biaser housing, and a spring extending within the biaser housing, and wherein the spring is configured to act on the biaser housing and the plate to contribute to the predetermined amount of force applied to the floor when the stabilizer is in the deployed position.

(7) Alternatively or additionally to any of the embodiments in this section, the assembly further comprises a force sensor configured to directly or indirectly measure an amount of force applied to the floor when the stabilizer is in the deployed position.

(8) Alternatively or additionally to any of the embodiments in this section, the predetermined amount of force applied to the floor when the stabilizer is in the deployed position is less than a force required to lift the surgical cart off of the floor.

(9) Alternatively or additionally to any of the embodiments in this section, the predetermined amount of force applied to the floor when the stabilizer is in the deployed position is within a range of 80 pounds of force to 100 pounds of force.

(10) Alternatively or additionally to any of the embodiments in this section, the stabilizer is a first stabilizer, the assembly further comprises a second stabilizer spaced from the first stabilizer, and the actuator is configured to simultaneously drive the first stabilizer and the second stabilizer toward the deployed position.

(11) Alternatively or additionally to any of the embodiments in this section, the assembly further comprises a mechanical release, wherein the mechanical release is configured to mechanically disengage the actuator and cause the stabilizer to move from the deployed position to the retracted position.

(12) Alternatively or additionally to any of the embodiments in this section, the actuator is configured to drive the stabilizer between the retracted position and the deployed position without regard to an amount of force being applied to the floor by the stabilizer.

(13) A further example may include a system comprising a first stabilizer and a second stabilizer

for stabilizing a surgical cart on wheels, a force applicator extending between the first stabilizer and the second stabilizer, and a linear actuator configured to pull down on the force applicator to cause the first and second stabilizers to move from a retracted position to a deployed position.

(14) Alternatively or additionally to any of the embodiments in this section, the force applicator includes a first end interacting with the first stabilizer, a second end interacting with the second stabilizer, and a portion between the first end and the second end that is coupled to the linear actuator.

(15) Alternatively or additionally to any of the embodiments in this section, the force applicator extends between the linear actuator, the first stabilizer, and the second stabilizer, and wherein the linear actuator is arranged to pull the force applicator toward a floor supporting the surgical cart to cause the first and second stabilizers to move toward the deployed position.

(16) Alternatively or additionally to any of the embodiments in this section, each of the first and second stabilizers comprise a foot configured to contact a floor to support the surgical cart when the stabilizer is in the deployed position, and a biaser configured to bias the stabilizer in the retracted position and contribute to an amount of force the foot applies to the floor in the deployed position.

(17) Alternatively or additionally to any of the embodiments in this section, the foot of each stabilizer is configured to initially translate with the force applicator and a biaser housing of the biaser when the stabilizer is moving from the retracted position to the deployed position and once the foot of each stabilizer contacts the floor, translate relative to the force applicator and the biaser housing.

(18) Alternatively or additionally to any of the embodiments in this section, wherein the linear actuator is configured to adjust between a first predetermined position associated with the retracted position and a second predetermined position associated with the deployed position irrespective of the amount of force the feet of the first and second stabilizers apply to the floor in the deployed position.

(19) Alternatively or additionally to any of the embodiments in this section, the system may further comprise a force sensor in communication with the linear actuator, the force sensor is configured to sense a force indicative of a force the first stabilizer and the second stabilizer apply to a floor contacting the first stabilizer and the second stabilizer when the first stabilizer and the second stabilizer are in the deployed position, and a controller configured to classify the deployment of the first stabilizer and the second stabilizer as being successful or unsuccessful based on an amount of force detected by the force sensor.

(20) A further example may include an assembly comprising a surgical cart for supporting a robotic arm and having locking casters, a first stabilizer supported by the surgical cart and having a first biaser and a first foot, a second stabilizer supported by the surgical cart and having a second biaser and a second foot, an actuator configured to adjust between a first position and a second position, and a force applicator configured to apply a force from the actuator to the first stabilizer and the second stabilizer. An adjustment of the actuator from the first position to the second position may move the first stabilizer and the second stabilizer from a retracted position to a deployed position. Each foot is configured to initially translate with the force applicator when the first stabilizer and the second stabilizer are moving from the retracted position to the deployed position and then, translate relative to the force applicator when the first stabilizer and the second stabilizer are moving from the retracted position to the deployed position. The first stabilizer and the second stabilizer are configured to stabilize the surgical cart when the first stabilizer and the second stabilizer are in the deployed position.

(21) Alternatively or additionally to any of the embodiments in this section, the actuator is a linear actuator configured to linearly adjust between the first position and the second position without regard to an amount of force being applied by the first stabilizer and the second stabilizer to a floor supporting the surgical cart.

(22) The above summary of some embodiments is not intended to describe each disclosed embodiment or every implementation of the present disclosure. The Figures, and Detailed Description, which follow, more particularly exemplify these embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The disclosure may be more completely understood in consideration of the following detailed description in connection with the accompanying drawings, in which:
(2) FIG. **1** is a schematic perspective view of an illustrative surgical cart;
(3) FIG. **2** is a schematic end view of the illustrative surgical cart of FIG. **1**;
(4) FIG. **3** is a schematic side view of the illustrative surgical cart of FIG. **1**;
(5) FIG. **4** is a schematic cross-sectional view of an illustrative cart base and stabilization system of the surgical cart of FIG. **1**, taken along line **4-4** in FIG. **3**;
(6) FIG. **5** is a schematic cross-sectional view of the illustrative cart base and stabilization system of the surgical cart of FIG. **1**, taken along line **5-5** in FIG. **2**;
(7) FIG. **6** the schematic cross-sectional view of FIG. **5**, in which an actuator is partially actuated;
(8) FIG. **7** the schematic cross-sectional view of FIG. **5**, in which the actuator is fully actuated; and
(9) FIG. **8** the schematic cross-sectional view of FIG. **5**, in which an emergency release is actuated.
(10) While the disclosure is amenable to various modifications and alternative forms, specifics thereof have been shown by way of example in the drawings and will be described in detail. It should be understood, however, that the intention is not to limit the invention to the particular embodiments described. On the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the disclosure.
DETAILED DESCRIPTION
(11) Carts may be used in or to facilitate surgical procedures. In some cases, carts may include one or more robots or robotic components for use in a surgical procedure. Such carts (e.g., robot carts or other suitable carts) often need to alternate between highly mobile states for transport and highly immobile states when the carts or components on the carts are being used in a surgical procedure or are otherwise being used. Often, locking casters are used to support the cart, which facilitate moving the cart in the highly mobile state, but can be insufficient, at least on their own, to fully stabilize the cart for use in surgical procedures in the highly immobile state.
(12) An assembly may include a cart and a stabilization system for use with the cart. The stabilization system may include feet that are deployed to apply a force to a surface or floor on which the cart is supported without lifting the cart off the surface or floor. In operation, a cart including the stabilization system and locking caster wheels may be moved to a desired location and once at the desired location, the locking caster wheels may be locked and feet of the stabilization system may deploy from beneath the cart or other suitable location.
(13) In one example implementation of the stabilization system, the stabilization system may include a first and second stabilizer each having a foot for contacting a floor, a crossbar (e.g., a force applicator), and an actuator. In operation, the actuator is activated in a first direction and pulls down on the crossbar acting on the stabilizers, which results in the stabilizers overcoming a bias force such that the feet are extended to the ground surface and apply force to the floor or another surface to stabilize the cart. To withdraw the feet from the floor, the actuator is released or activated in a second direction opposite of the first direction such that a bias force of the stabilizers overcomes a force, if any, acting thereon by the crossbar and causes the feet to withdraw from the surface.
(14) The amount of force applied to the surface by the feet may be any suitable amount of force that is able to stabilize the cart. The force may, but need not, be sufficient lift the cart (e.g., casters

or other suitable portions of the cart) off of the surface on which the cart is supported. In one example, an amount of forced applied to a surface supporting the cart by the stabilizers may be in the range of about eighty pounds of force to about one-hundred pounds of force. Other suitable force amounts are contemplated.

(15) Turning to the Figures, FIG. **1** depicts a schematic perspective view of an illustrative cart assembly **100**. The cart assembly **100** may include a cart **102** and a stabilizing system **104**, among other suitable components.

(16) The cart **102** may be a cart that is configured for use in surgical procedures. In one example, the cart **102** may include or may be configured to receive a surgical robot (e.g., robotic arms, etc.) or other suitable teleoperated surgical systems.

(17) The cart **102** may include a frame structure **106**. The frame structure **106** may have any suitable design or configuration that facilitates supporting one or more items for use in a surgical procedure, such as one or more computer systems, cameras, lights, instruments, surgical robots, suitable teleoperated surgical systems, related components, or combinations thereof. In some cases, the frame structure **106** may include or otherwise be supported by a base **130** (e.g., a surgical cart base).

(18) As depicted in FIG. **1**, the frame structure **106** may have or define a top **108**, a bottom **110**, and one or more sides **112** of the cart, where the one or more sides **112** may extend entirely or at least partially between the top **108** and the bottom **110**. In some cases, the top **108**, the bottom **110**, and the one or more sides **112** may define a central space **114** in which one or more components of or used with the cart assembly **100** may be located. The frame structure **106** may include struts, frame bars, supports, or other suitable components to support defining the top **108**, the bottom **110**, and the sides **112**, along with components connected thereto or supported thereby. Although not depicted, one or more of the top **108**, the bottom **110**, and the sides **112** may include or have thereon a cover or other suitable solid or partially solid surface.

(19) The top **108** of the cart **102** may be configured to support one or more surgical system components or other components. Although not required, the top **108** may have one or more components configured to facilitate securing the surgical system components or other components to the cart **102**. An example surgical system component secured to the cart **102** (e.g., secured to the top **108** of the cart **102** or other suitable location) may be a robotic surgical arm, thought other components are contemplated.

(20) The bottom **110** of the cart **102** may be configured to support one or more components located at least partially within the central space **114**. In one example, as depicted in FIG. **1**, the stabilizer system **104** may be supported by the bottom **110** of the cart **102**. Such a configuration is not required, and the stabilizer system **104** may be supported by one or more other portions of the cart **102**.

(21) The frame structure **106** may support or define other features of the cart **102**. As depicted in FIG. **1**, the frame structure **106** may include one or more monitor supports **116** and one or more handles **118**, among other suitable features. The monitor support **116** may be configured to receive a display or control system configured to be used with (e.g., to facilitate control of) components supported by the cart **102** or surgical components. In some cases, the handle **118** may be configured as a rigid handle or the handle **118** may have an adjustable portion that may be adjusted to lock or unlock wheels of the cart **102**, but this is not required. Although the monitor support **116** and the handle **118** are described herein as being part of the frame structure **106**, the monitor support **116** or the handle **118** may be separate from the frame structure **106** and supported by the frame structure **106**.

(22) The cart **102** may include one or more wheels **120** that may be configured to facilitate supporting the cart **102** on a surface or floor and facilitate movement of the cart **102** on or along the surface or floor. In one example, the cart may include four wheels **120** (though, only three are visible in FIG. **1**).

(23) The wheels **120** may be of any suitable type or suitable combination of types configured to support the cart **102** and facilitate transporting the cart along a floor or other suitable surface. Example types of wheels **120** include, but are not limited to, rigid caster wheels, swivel caster wheels, locking wheels, locking caster wheels, kingpinless caster wheels, omnidirectional wheels, etc. In one example, the cart **102** may include one or more wheels **120** that are of the locking caster type.
(24) When cart **102** includes one or more locking wheels **120** (e.g., locking caster wheels, etc.), wheels **120** may be configured to manually lock or unlock. In one example, the wheels **120** may be manually locked at each individual wheel **120**. In another example, one or more of the wheels **120** may be manually locked or unlocked by a user actuating a mechanical locking system that locks-in-place or unlocks one or more of the wheels **120**.
(25) Alternatively or additionally, the wheels **120** may be configured to lock or unlock in an automated manner in response to an action by a user or a positioning of a cart at a location in a room. In one example, the wheels **120** may be locked-in-place or unlocked in response to a user selecting, or a control system receiving a selection of, a button or other suitable actuator that initiates an electromechanical system to lock or unlock one or more of the wheels **120**. In another example, the wheels **120** may be locked-in-place or unlocked in response to a positioning system in a room (e.g., surgery room or other suitable room) identifying the cart **102** is at a desired or predetermined location within the room or with respect to a feature in the room and initiating an electromechanical system to lock or unlock one or more of the wheels in response to identifying the position of the cart **102**.
(26) As discussed above, the cart assembly **100** may include the stabilizer system **104**. Among other features, and as discussed in greater detail below, the stabilizer system **104** may include one or more stabilizers **122** having a foot **123** or other suitable feature for engaging a floor or other surface to facilitate stabilizing the cart **102**, one or more actuators **124** configured to drive the stabilizer(s) **122** toward a deployed position or a retracted position, and one or more force applicator **126** (e.g., a crossbar or other suitable force applicator) coupling the one or more actuators **124** and the one or more stabilizers **122**. In one example, the stabilizer system **104** may include two stabilizers **122**.
(27) In operation, when the one or more actuators **124** are actuated, the actuator(s) may drive the stabilizer(s) **122** from an undeployed state (e.g., a retracted or resting position) to a fully-deployed state or at least partially deployed state causing the feet **123** to move toward or to a floor or other suitable surface. When two or more stabilizers **122** are utilized with the force applicator **126**, actuating the actuator **124** may simultaneously drive the stabilizers **122** (e.g., via the force applicator **126**) toward the deployed position. Simultaneously drive of the stabilizers **122**, however, is not required and the configuration of the stabilizers **122**, the actuator(s) **124**, and the force applicator(s) **126** may be alternatively or additionally configured such that one or more of the stabilizers **122** are actuated separate or independent from one or more of the stabilizers **122**. When in a fully deployed state such that the feet **123** are in contact with the floor or other suitable surface and applying a force thereto, the feet **123** and the stabilizer(s) **122** may resist movement of, or otherwise stabilize, the surgical cart **102**.
(28) The actuator(s) **124** may be manually actuated or actuated in an automated manner. In some cases, actuation of the actuator **124** may be performed automatically in response to or at the time of locking the wheels **120**, but this is not required.
(29) The cart **102** may include an emergency release **128** (e.g., a mechanical release). In some cases, the emergency release **128** may be coupled to the locks or locking system of the wheels **120**, part of or coupled to the stabilizer system **104**, or coupled to both of the locks or locking system of the wheels **120** and the stabilizer system **104**. When the emergency release **128** is coupled to the locks or locking system of the wheels **120** and the stabilizer system **104** and the emergency release **128** is actuated, the locks or locking system of the wheels **120** may cause the wheels **120** to be in an

unlocked state and the stabilizer system **104** may cause the stabilizers **122** to move the feet **123** to a fully or at least partially withdrawn or retracted position such that the cart **102** may be moved along a floor or other suitable surface.
(30) FIG. **2** depicts a schematic end view of the cart assembly **100**. FIGS. **5-8** depict cross-sectional views of the cart assembly **100** taken along line **5-5** in FIG. **2**.
(31) The schematic end view shown in FIG. **2** depicts an end of the cart **102** at which the monitor support **116** and the handle **118** are located. As can be seen in FIG. **2**, the cart assembly **100** may include four wheels **120**, where the front wheels **120** may be slightly offset (e.g., inward) from the back wheels **120**, but this is not required and the wheels **120** may be offset in a different manner or not offset at all. Further, the stabilizer system **104** may be centrally located within the cart **102** between opposite sides **112**. The stabilizer system **104** centrally located in the cart **102** may facilitate providing well-balanced stabilization of the cart **102** when the stabilizers **122** are in a deployed position. Although the stabilizer system **104** is centrally located within the central space **114** in the cart assembly **100** configuration depicted in FIG. **2**, the stabilizer system **104** may be located elsewhere with respect to cart **102**.
(32) FIG. **3** depicts a schematic side view of the cart assembly **100**. FIG. **4** depicts a cross-sectional view of the cart assembly **100** taken along line **4-4** in FIG. **3**.
(33) As can be seen in FIG. **3** due to only depicting a front wheel **120** and a back wheel **120**, the front wheels **120** are axially aligned and the back wheels **120** are axially aligned. Further, the stabilizer system **104** may include a first stabilizer **122***a* with an associated first foot **123***a* and a second stabilizer **122***b* with an associated second foot **123***b*. As depicted, the first stabilizer **122***a* may be located closer to a first end (e.g., a back end) of the cart **102** and the second stabilizer **122***b* may be located closer to a second end (e.g., a front end) of the cart **102**. The stabilizers **122** may be spaced from the ends of the cart **102** as desired to facilitate stabilizing the cart when in use.
(34) FIG. **4** depicts a schematic cross-sectional view of the cart assembly **100**. The cross-sectional view of the cart assembly **100** is taken along line **4-4** in FIG. **3**, showing the actuator **124** coupled to the force applicator **126** and the base **130** of the cart **102** with the frame structure **106** removed.
(35) As discussed above, the force applicator **126** may be coupled to the actuator **124** and the stabilizer(s) **122** such that the actuator **124** may act on the force applicator **126** to cause the force applicator **126** to apply a force to the stabilizer(s) **122**. The actuator **124** may be coupled to the force applicator **126** in any suitable manner that facilitates the actuator **124** acting on the force applicator **126**. Example couplings between the force applicator **126** and the actuator **124** may include, but are not limited to, a pin connection, a threaded connection, a luer lock connection, a bayonet connection, a friction fit connection, a snap connection, other suitable couplings, or combinations thereof.
(36) In one example configuration of the coupling between the force applicator **126** and the actuator **124**, as depicted in FIG. **4**, the actuator **124** may be coupled to the force applicator **126** with a pin **432** to create a pin connection. In the example, the pin **432** may be inserted into an opening **434** extending through the actuator **124** and the actuator **124** with the pin **432** may be inserted into the force applicator **126**. In such a configuration, when the actuator **124** is adjusted or otherwise actuated, the pin may act on the force applicator **126** to cause the force applicator **126** to change position. The pin **432** may be part of the actuator **124** or a separate component used with the actuator **124**.
(37) The force applicator **126** may have any suitable configuration. Example configurations of the force applicator **126** may include a single elongated component or bar, a multi-piece component or bar, or one or more other suitable configuration. In one example, as depicted in FIG. **4**, the force applicator may include a first elongated component **436** and a second elongated component **438**. Although not required, the first elongated component **436** may have a bar configuration having an opening **440** for receiving the actuator **124** and the pin **432**. The second elongated component **438** may have a U-bracket configuration that is configured to receive and couple to (e.g., via any

suitable connectors) the first elongated component **436**, but this is not required and other configurations are contemplated. Further, the second elongated component **438** may have an opening **442** configured to receive the actuator **124**.
(38) The force applicator **126** may be formed from any suitable material configured to withstand forces applied to it in use with the actuator **124** and the stabilizer(s) **122**. Example materials include, but are not limited to, metals, polymers, stainless steel, sheet metal, combination of metals and polymers, etc.
(39) The actuator **124** may be coupled to the base **130** in any suitable manner. In the example depicted in FIG. **4**, the actuator **124** may extend through an opening **444** in the base **130** and may be coupled to a connector plate **446** that is secured to the base **130**. In the example depicted in FIG. **4**, a pin **448** may extend through a clevis **450** to connect the actuator **124** to the connector plate **446**. Other suitable configurations are contemplated. Any suitable types of couplings or connectors may be utilized to couple or connect the actuator **124**, the base **130**, and the connector plate **446** including, but not limited to, screws, bolts, nuts, washers, spacers, pins, etc.
(40) The actuator **124** may be any suitable type of actuator configured to apply a pulling or pushing force on the force applicator **126** and in-turn the stabilizer(s) **122**. Example types of actuators include, but are not limited to, linear actuators, rotary actuators, hydraulic actuators, pneumatic actuators, electric actuators, thermal magnetic actuators, mechanical actuators, supercoiled polymer actuators, or other suitable actuators and combinations of actuator types. In one example, the actuator **124** may be an electric motor driven linear actuator, but this is not required.
(41) The electric motor linear actuator **124** depicted in FIG. **4** may have a variety of components. As depicted, the actuator **124** may include an outer tube **452** (e.g., a cover tube), an inner tube **454** (e.g., an extension tube, drive tube, translating tube, etc.), an electric motor **456**, and a motor housing **458**. The actuator **124** may include other components not depicted that include, but are not limited to, a spindle or lead screw, a safety stop located on the end of the spindle to prevent over extension of the inner tube, a wiper between the outer tube **452** and the inner tube **454**, a drive nut attached to the inner tube **454** and travels along the spindle, limit switch to control the fully extended and retracted inner tube position by cutting current to the motor, a gear, etc.
(42) FIG. **5** depicts a schematic cross-sectional view of the cart assembly **100**. The cross-sectional view of the cart assembly **100** taken along line **5-5** in FIG. **2**, shows, among other features, the force applicator **126** coupled to the actuator **124** and the stabilizers **122** (e.g., a first stabilizer **122***a* and a second stabilizer **122***b*).
(43) The stabilization system **104** may include one or more pressure or force sensors **560** for determining an amount of force the stabilizers **122** are applying to a floor or surface **561** supporting the cart **102**. The force sensor(s) **560** may be configured to directly, indirectly, or both indirectly and directly measure or sense the amount of force applied to or indicative of the amount of force applied to the floor or surface **561** when the stabilizer(s) **122** are in a deployed position. A sensed or measured force may be utilized for any suitable purpose including, but not limited to, as a check on the actuator **124** to ensure the actuator **124** is operating at specifications. If it is determined that the actuator **124** is not operating at specifications, data from the force sensor **560** may be utilized to determine the health of the actuator **124** and actions to take for addressing the actuator **124** not operating at specifications. Additionally or alternatively, the output of the force sensor(s) **560** may be utilized (e.g., by a controller of the cart assembly **100**, or other suitable controller, receiving an output signal directly or indirectly from the force sensor **560**) to classify a deployment of the stabilizer(s) **122** as being successful or unsuccessful based on an amount of force detected by the force sensor **560**. In response to detecting an unsuccessfully deployment, the controller may provide a notification, take a corrective action, make a second deployment attempt, or take other suitable action, but this is not required.
(44) The force sensors **560** may be located at any suitable location(s) along the stabilizer system **104** at which an amount of force applied to or indicative of the amount of force applied to the floor

or surface **561** when the stabilizer(s) **122** are in the deployed position may be measured or sensed. In one example when the force sensor **560** is configured to directly measure the force applied to the floor or surface **561**, the force sensor **560** may be located in the foot **123** of the stabilizer **122** such that one portion of the force sensor **560** is configured to contact the floor or surface **561** and another portion of the force sensor **560** is configured to contact the stabilizer **122**. In another example, when the force sensor **560** is configured to indirectly measure the force applied to the floor or surface, the force sensor **560** may be spaced away from the foot **123** that is applying force to the floor or surface **561**, or otherwise configured so as not to be in direct contact with the floor or surface **561**.

(45) In some cases, the force sensor **560** may be secured to the base **130** of the cart **102** and the connector plate **446** to which the actuator **124** is coupled. In such a configuration, when the actuator **124** is acting on the force applicator **126** and the stabilizer(s) **122** in a first direction represented by arrow D**1**, the actuator **124** applies a force to the connector plate **446**, which in turn applies a force between the force sensor **560** and the base **130**. The force between the force sensor **560** and the base **130** is an indirect measure of the amount of force applied to the floor or other surface when the stabilizer(s) **122** are in the deployed position. Other configurations are contemplated.

(46) The pressure or force sensors **560** may be of any suitable type. Example pressure or force sensor types include, but are not limited to, load cells, strain gages, for sensing resistors, optical force sensors, ultrasonic force sensors, pneumatic load cells, hydraulic load cells, piezoelectric crystal load cells, inductive load cells, capacitive load cells, magnetoresistive load cells, strain gage load cells, other types, or combinations thereof.

(47) When the stabilizer system **104** includes two stabilizers **122**, the actuator **124**, and the force applicator **126**, the two stabilizers **122** may be located at any desired location with respect to the actuator **124** and the force applicator **126** may extend between the two stabilizers **122** and the actuator **124**. As depicted in FIG. **5**, the stabilizers **122** may be spaced an equal distance or substantially equal distance from the actuator **124**. Such a configuration of the stabilizers **122** with respect to the actuator **124** may facilitate applying a substantially equal force to the floor or other surface **561** supporting the cart **102** at each of the stabilizers **122**. For example, the actuator **124** may be configured to pull down on the force applicator **126** (e.g., pull the force applicator **126** toward the floor or other surface **561**) and cause the two stabilizers **122** to move from a resting or retracted position (e.g., the undeployed position) to a deployed position and apply equal or substantially equal force to the floor or other surface **561**.

(48) The stabilizers **122** may be coupled to the actuator **124** via the force applicator **126** such that the actuator **124** may act on the force applicator **126** to cause the force applicator **126** to apply a force to the stabilizers **122**. The stabilizers **122** may be coupled to the force applicator **126** in any suitable manner. Example couplings between the force applicator **126** and the stabilizers **122** may include, but are not limited to, a pin connection, a locking pin connection, a threaded connection, a luer lock connection, a bayonet connection, a friction fit connection, a snap connection, or other suitable couplings.

(49) The stabilizers **122** and the actuator **124** may be coupled to the force applicator **126** at any suitable location. In one example, the first stabilizer **122***a* may be coupled to or interact with a first end of the force applicator **126**, the second stabilizer **122***b* may be coupled to or interact with a second end of the force applicator **126**, and the actuator **124** may be coupled to or interact with a portion of the force applicator **126** between the first end and the second end, but other configurations are contemplated.

(50) In one example configuration of the coupling between the force applicator **126** and the stabilizers **122**, as depicted in FIG. **5**, the stabilizers **122** may be coupled to the force applicator **126** with a connector **562** (e.g., a clip, a retaining clip, a pin, a screw, a bolt, a nut, bracket, etc., which may be a component of the stabilizers **122**) extending around a portion of the stabilizers **122**. In the

example, a portion of a biaser housing **568** of the stabilizer **122** may be inserted into the force applicator **126** and the connector **562** may be applied through, around, or otherwise connected to the portion of the biaser housing **568** inserted into the force applicator **126** to couple the stabilizer **122** with the force applicator **126**. In such a configuration, when the actuator **124** is adjusted or otherwise actuated, the force applicator **126** may act on an outer surface of the stabilizers **122** to cause the stabilizer to move between retracted and deployed positions.
(51) The stabilizers **122** may have any suitable configuration. In some cases, the stabilizers **122** may be configured to apply a predetermined amount of force to the floor or surface **561** in response to the actuator **124** acting thereon. Although not required, a predetermined amount of force applied to the floor or surface **561** may be an amount of force applied to the floor or surface **561** when the stabilizer **122** is in the deployed that is less than an amount of force required to lift the cart **102** off of the floor, but provides enough force on the floor or surface **561** to stabilize the cart **102** and resist, prevent, or substantially prevent the cart **102** from moving when the stabilizers **122** are in the deployed position. Although other force amounts are contemplated and may depend on a mass or other configuration of the cart **102**, an example predetermined amount of force applied to the surface or floor **561** when the stabilizers **122** are in the deployed position without lifting the cart **102** may be a force amount in a range of eighty or about eighty pounds of force to one-hundred or about one-hundred pounds of force. Other suitable amounts of force are contemplated.
(52) In one example configuration, as depicted in FIG. **5**, the stabilizers **122** may include one or more feet **123** configured to contact the floor or other surface **561** to support the surgical cart **102** when the stabilizers **122** are in the deployed position, the rod **564** that may be in communication with the feet **123**, a support or plate **574** that may be fixed to or fixed relative to the rod **564** and may translate with the rod **564** (e.g., the support or plate **574** and the rod may axially adjust within a biaser housing **568**) and the foot **123**, and a biaser **566**. In some cases, the biaser **566** may be configured to bias the foot **123** in a retracted position and contribute to a predetermined amount of force applied to the floor or surface **561** supporting the cart **102** when the foot **123** is in a deployed position. Alternatively, the biaser **566** may be configured to bias the foot **123** in the deployed position.
(53) The feet **123** may take on any suitable configuration. In some cases, the feet **123** may have a rounded, a squared, a circular, or other suitable configuration for supporting and stabilizing the cart **102** when the stabilizers **122** are in the deployed configuration. The feet **123** can be constructed from any of a variety of materials. In an example, the feet **123** are constructed from plastic.
(54) The foot **123** may be coupled to or otherwise connected to the rod **564** in any suitable manner. In one example, the foot **123** may be rigidly coupled to the rod **564** such that the foot **123** cannot move independent of the rod **564**. In another example, the foot **123** may be pivotally or otherwise adjustably coupled to the rod **564**. When pivotally or otherwise adjustably coupled to the rod **564**, the foot **123** may be configured to facilitate stabilizing the cart on an uneven floor or surface.
(55) The rods **564** may take on any suitable configuration. As depicted in FIG. **5**, the rods **564** may have a first portion **564***a* and a second portion **564***b*, but this is not required and the rod **564** may have a single portion or more than two portions. When one or more of the rods **564** have more than one portions, the rods **564** may be connected in any suitable manner including, but not limited to, weld connections, threaded connections, adhesive connections, over molding connections, etc.
(56) The support or plate **574** may take on any suitable configuration. In one example, the support or plate **574** be configured to translate within the biaser housing **568**, further discussed below, and may extend around the rod **564** such that the plate **574** may support an end of a second spring (e.g., a second spring **572**, further discussed below) of the biaser **566**.
(57) Further, in some cases, the stabilizers **122** may include one or more supports **576** extending toward the floor or other surface **561** from the base **130** of the cart **102** and the rods **564** may extend through the one or more supports **576**. The supports **576**, when included, may facilitate structurally supporting the feet **123** adjacent the support **576** or the rods **564** extending through the

supports.

(58) The supports **576** may include adjustable bearings **580** (e.g., bearings made of metal, rubber, etc.) that may extend from the supports **576** to support the feet **123**. Any suitable number of bearings may be utilized. For example, a single bearing may be utilized, two bearings may be utilized, three bearings may be utilized, ten bearings may be utilized, etc. Further the bearings **580** may take on any suitable shape configured to support or otherwise contact the feet **123** when the feet **123** are in the resting or retracted position (e.g., a non-stabilizing position). Example shapes include, but are not limited to, full-circle shapes, half-circle shapes, ornamental shapes, ring shapes, dot shapes, etc. As depicted in the Figures, three bearings are utilized and each bearing has a rounded dot configuration with an elongated threaded portion for coupling to the support **576**. Other configurations are contemplated for coupling the bearings **580** to the supports **576**. Additionally or alternatively to the bearings **580** being of or otherwise coupled to the supports **576**, the bearings may be of or coupled to the feet **123**.

(59) The biaser **566** may include one or more components configured to bias the foot **123** and the stabilizer **122** in the retracted position and contribute to a predetermined amount of force applied to the floor or surface **561** supporting the cart **102** when the foot **123** and the stabilizer **122** are in the deployed position. Among other components, the biaser **566** may include the biaser housing **568**, a first spring **570** biasing the stabilizer **122** to a retracted or resting position, and a second spring **572** biasing the stabilizer **122** to a deployed position, as depicted in FIG. **5**.

(60) The biaser housing **568** may take on any suitable configuration. In some cases, the biaser housing **568** may have a first portion **568***a* and a second portion **568***b*. As depicted in FIG. **5**, first portion **568***a* of the housing **568** may be configured to be received within the second portion **568***b* of the housing **568**. The first portion **568***a* and the second portion **568***b* of the biaser housing **568** may be fixed with respect to one another. Alternatively, the first portion **568***a* and the second portion **568***b* may be configured to slide or otherwise adjust with respect to one another in response to external, internal, or both external and internal forces acting on the stabilizer **122**. Other configurations of the housing **568** are contemplated.

(61) The first spring **570** of the biaser **566** may be located at any suitable location with respect to the biaser housing **568**. In one example, the first spring **570** of the biaser **566** may be entirely or at least partially exterior of the biaser housing **568** (e.g., the first spring **570** may extend along an exterior of the biaser housing **568**) and configured to bias the stabilizer **122** to a retracted position by acting on the biaser housing **568** in a direction of arrow D**2** (e.g., away from the surface **561** in a direction substantially perpendicular to the surface **561**). Although not required, the first spring **570** may extend between a surface of the base **130** of the cart **102** and the second portion **568***b* of the housing **568**. In such a configuration, the first spring **570** may act on the base **130** and the biaser housing **568** to bias the stabilizer **122** to the retracted or resting position.

(62) The second spring **572** of the biaser **566** may be located at any suitable location with respect to the biaser housing **568**. In one example, the second spring **572** of the biaser **566** may be entirely or at least partially within an interior of the biaser housing **568** (e.g., the second spring **572** may extend along an interior of the biaser housing **568**) and configured to bias the stabilizer **122** to a deployed position by acting on the support or plate **574** in a direction of arrow D**1** (e.g., toward from the surface **561** in a direction substantially perpendicular to the surface **561**). Although not required, the second spring **572** may extend between the second portion **568***b* (e.g., an interior surface of the second portion **568***b*) of the biaser housing **568** and the first portion **568***a* of the biaser housing **568** or the plate **574** extending around the rod **564** within the biaser housing **568**. In a configuration where the second spring **572** extends between the second portion **568***b* of the biaser housing **568** and the support or plate **574**, the second spring **572** may act on the biaser housing **568** and the support or plate **574** to contribute to a predetermined amount of force applied to the floor or other surface **561** stabilizing the cart **102** when the stabilizer **122** is in the deployed position.

(63) The first spring **570** and second spring **572** may have any suitable shape or configuration. In

one example, the second spring **572** may have a smaller diameter than the first spring **570**, the second spring **572** have a longer relaxed state than the first spring **570**, and the first spring **570** and the second spring **572** may have one or more other similar or different configurations, but this is not required.

(64) The first and second springs **570**, **572** may each have any suitable spring constant. The spring constants may be determined, selected, or set based on one or more of a variety of factors, including, but not limited to, a mass of equipment on the cart **102**, a mass of the cart **102**, desired relative spring constants between the first spring **570** and the second spring **572**, an amount of force the actuator **124** may be able to overcome, a desired amount of force to be applied to the floor or surface **561** by the stabilizers **122**, etc. In one example, a spring constant of the first spring **570** may be less than a spring constant of the second spring **572** such that the first spring **570** contributes to an amount of force needed (e.g., an amount of force the actuator **124** applies to the force applicator **126**) to advance the stabilizer **122** from the retracted position to a position at which the stabilizer **122** contacts the floor or surface **561**. The spring constant of the second spring **572** may be greater than a spring constant of the first spring **570** such that the spring constant of the second spring **572** may contribute to an amount of force applied to the floor or surface **561** by the stabilizer **122** in a deployed configuration after the stabilizer **122** contacts the floor or surface **561**.

(65) As depicted in FIG. **5**, when the stabilizers **122** are in a retracted or resting position, the first springs **570** are expanded between the base **130** and second portion **568***b* of the biaser housing **568** and have a length S**1**. In the retracted or resting position of the stabilizers **122**, the second springs **572** extend between the second portion **568***b* of the biaser housing **568** and the plate **574** and have a length S**2**. The actuator **124** has a length of S**3** when the stabilizers **122** are in the retracted or resting position.

(66) The stabilizers **122** may transition from the retracted or resting position to a deployed position in response to actuation of the actuator **124**, as discussed below with respect to FIGS. **6** and **7**. Other suitable configurations are contemplated for transitioning between the retracted or resting position and the deployed positions.

(67) FIGS. **6** and **7** depict the transition of the stabilizers **122** of the cart from a retracted or resting position (e.g., as depicted in FIG. **5**) to a fully deployed position (e.g., as depicted in FIG. **7**). The stabilizers **122** transition from the retracted or resting position to the fully deployed position in response to adjustment of the actuator **124**, but other configurations are contemplated.

(68) The cross-sectional view of the cart assembly **100** in FIG. **6** depicts, among other features, the stabilizers **122** in a transitional position between the retracted or resting position and the fully deployed position at which the actuator **124** has been adjusted from a length of S**3** (as in FIG. **5**) to a length of S**3**′ that is shorter than the length S**3**. In adjusting the actuator **124**, the actuator **124** pulls down on the force applicator **126** and the stabilizers **122** toward the floor or other surface **561** in a direction of arrow D**1**. As the stabilizers **122** are moving from the retracted position to the deployed position, each foot **123** of the stabilizers **122** may initially translate with the force applicator **126** and the biaser housing **568**. That is, in adjusting to the transitional position, the actuator **124** at least partially overcomes the bias force of the first spring **570** and causes the first spring **570** to have a reduced length S**1**′ that is shorter than the length S**1**. At this point, the second spring **572** has a length of S**2**′ that is equal to or substantially equal to the length S**2**, and the feet **123** are at a partially deployed state at which the feet **123** are spaced from the floor or other surface **561**.

(69) The cross-sectional view of the cart assembly **100** in FIG. **7** depicts, among other features, the stabilizers **122** in the fully deployed position at which the actuator **124** has been adjusted to a desired position, such that a length of the actuator **124** has changed from a length of S**3** (as in FIG. **5**) to a length of S**3**″ that is shorter than the lengths S**3** and S**3**′. In some cases, the position of the actuator **124** in the fully deployed position may be a predetermined linear location that is configured, in conjunction with a configuration of the stabilizers, to cause a known amount of force

to be applied to the floor or other surface **561**. Such deployment can be without respect to a force sensor or other sensor and can simplify the system. Alternatively or additionally, the position of the actuator **124** may be determined in response to measured or sensed values of force applied to the floor or other surface **561** by the feet **123** as the actuator **124** is adjusted. For example, the actuator **124** may be adjusted until a desired (e.g., predetermined or otherwise) amount of force is applied to the floor or other surface **561** by the feet **123**, but this is not required. Driving the actuator **124** to a predetermined linear location to cause a known amount of force to be applied to the floor or other surface **561** may simplify a control of the stabilizer system **104** over other stabilizing systems that may utilize a controller to adjust an actuator until a desired stabilizing force is applied to a supporting surface.

(70) In adjusting to the fully-deployed position, the actuator **124** may overcome the bias force of the first spring **570** and when at the predetermined linear location, cause the first spring **570** to have a further reduced length S**1**″ that is shorter than the length S**1** and S**1**′ and at which the foot **123** of the stabilizer **122** contacts the floor or other surface **561**. Once the foot **123** of the stabilizer **122** contacts the floor or other surface **561**, the foot **123** or one or more of the components indirectly or directly coupled to the foot **123**, such as the rod **564**, the support or plate **574**, the second spring **572**, or a combination thereof, may adjust (e.g., translate) relative to the force applicator **126** and the biaser housing **568**, as depicted in FIG. **7**, until the actuator **124** is at the predetermined linear position and the predetermined amount of force is applied to the floor or other surface **561** by the foot **123**. In an example, the rod **564**, the support plate **574**, and the second spring **572** may adjust relative to the force applicator **126** and the biaser housing **568** until the predetermined amount of force is applied to the floor other surface **561**. That is, once the feet **123** are contacting the floor or other surface **561** and the actuator **124** is at the predetermined linear position, the second spring **572** may compress to a length of S**2**″ that is shorter than the lengths S**2** and S**2**′ and at which the feet **123** are in contact with and applying a desired amount of force to the floor or other surface **561** to stabilize the cart **102** at a desired location.

(71) In some cases, the desired or predetermined amount of force applied to the floor or surface **561** may be entirely or at least partially dependent on the spring constant of the second spring **572**. In one example, the predetermined amount of force applied to the floor or surface **561** may be dependent on the spring constant of the second spring and a distance the actuator **124** travels after the feet **123** contact the floor or surface **561**, but this is not required.

(72) As discussed, the actuator **124** may be adjusted to move the stabilizers **122** between the retracted or resting position and the deployed position. In some cases, the actuator **124** may be configured to adjust between a first predetermined position associated with the resting or retracted position (e.g., the undeployed position) and a second predetermined position associated with the deployed position of the stabilizers **122**. The second predetermined position of the actuator **124** that is associated with the deployed position of the stabilizers **122** may be a location independent of or irrespective of an amount of force the feet **123** of the stabilizers **122** apply to the floor or other surface **561** in the deployed position. For example, the second predetermined position of the actuator **124** may be set to cause the feet **123** of the stabilizers **122** to contact the floor or surface **561** and once the feet **123** are in contact with the floor or surface **561**, the force caused by the spring constant of the second spring **572** may cause the feet **123** to apply the predetermined amount of force to floor or other surface **561**. That is, the actuator **124** may drive the stabilizer **122** between the retracted position and the deployed position without regard to an amount of force being applied to the floor by the stabilizer **122**. Such a configuration of the actuator **124** relative to the stabilizer **122** may allow for a simplified control of the actuator **124** and the stabilizer **122** due to only needing to adjust the actuator **124** to a single, known or predetermined position each time the stabilizer **122** is to be deployed.

(73) The actuator **124** may be adjusted manually (e.g., in response to adjustment of a lever or other mechanical mechanism) or in an automated manner in response to a control signal manually

initiated by a user selection or automatically initiated. In some cases, however, it may be desirable to quickly release the stabilizers from the deployed position (e.g., to facilitate moving the cart away from a surgical site in response to an emergency or for one or more other suitable purposes). To facilitate a quick or emergency release of the stabilizers **122**, the emergency release **128** may be actuated. In one example, the emergency release **128** may be a mechanical release that is actuated from a first position with the stabilizers **122** in the fully-deployed position (e.g., as depicted in FIG. **7**) to a second position (e.g., as depicted in FIG. **8**) that causes or otherwise results in the stabilizers **122** being retracted in the second direction of arrow D**2** to the retracted position. In the configuration depicted in FIG. **8**, the emergency release **128** may be rotated in the direction of arrow A, however, other configurations of the emergency release are contemplated including, but not limited, to a push button.
(74) The emergency release **128** may be adjusted manually (e.g., in response to adjustment of a lever or other mechanical mechanism), may be adjusted in an automated manner in response to a control signal automatically initiated or manually initiated by a user selection, or may be configured to be adjusted manually or in an automated manner. When configured to be adjusted manually, the emergency release **128** may be include a grip portion **578** that may be engaged by a user's hand or foot or other suitable component to facilitate actuating the emergency release **128**.
(75) The emergency release **128** may be connected or coupled to the actuator **124** or other component of the stabilizers **122** that may facilitate moving the stabilizers **122** to a retracted position in any suitable manner. In one example, the emergency release **128** may be mechanically, electrically, or both mechanically and electrically connected or coupled to actuator **124** such that actuation of the emergency release **128** causes the actuator **124** or other suitable component of the stabilizer system **104** to mechanically release from an actuated position at which the actuator **124** has the length S**3**″ to a relaxed position, causing the stabilizer(s) **122** to automatically move from the deployed position at which the first spring has the length S**1**″ and the second spring has the length S**2**″ to the retracted position, as depicted in FIG. **8**, such that the first spring **570** has the length S**1**, the second spring **572** has the length S**2**, and the actuator has the length S**3**.
(76) For the following defined terms, these definitions shall be applied, unless a different definition is given in the claims or elsewhere in this specification.
(77) All numeric values are herein assumed to be modified by the term “about”, whether or not explicitly indicated. The term “about” generally refers to a range of numbers that one of skill in the art would consider equivalent to the recited value (e.g., having the same function or result). In many instances, the term “about” may include numbers that are rounded to the nearest significant figure.
(78) The recitation of numerical ranges by endpoints includes all numbers within that range (e.g. 1 to 5 includes 1, 1.5, 2, 2.75, 3, 3.80, 4, and 5).
(79) As used in this specification and the appended claims, the singular forms “a”, “an”, and “the” include plural referents unless the content clearly dictates otherwise. As used in this specification and the appended claims, the term “or” is generally employed in its sense including “and/or” unless the content clearly dictates otherwise.
(80) It is noted that references in the specification to “an embodiment”, “some embodiments”, “other embodiments”, etc., indicate that the embodiment described may include one or more particular features, structures, and/or characteristics. However, such recitations do not necessarily mean that all embodiments include the particular features, structures, and/or characteristics. Additionally, when particular features, structures, and/or characteristics are described in connection with one embodiment, it should be understood that such features, structures, and/or characteristics may also be used connection with other embodiments whether or not explicitly described unless clearly stated to the contrary.
(81) The above detailed description should be read with reference to the drawings in which similar elements in different drawings are numbered similarly. The drawings, which are not necessarily to

scale, depict illustrative embodiments and are not intended to limit the scope of the invention.
(82) It should be understood that this disclosure is, in many respects, only illustrative. Changes may be made in details, particularly in matters of shape, size, and arrangement of steps without exceeding the scope of the disclosure. This may include, to the extent that it is appropriate, the use of any of the features of one example embodiment being used in other embodiments. The invention's scope is, of course, defined in the language in which the appended claims are expressed.

Claims

1. An assembly comprising: a surgical cart having locking caster wheels configured to contact a floor; a stabilizer comprising: a foot configured to resist movement of the surgical cart when the foot is in a deployed position contacting the floor; and a biaser configured to bias the foot in a retracted position and contribute to a predetermined amount of force applied to the floor when the foot is in the deployed position, the biaser comprising a biaser housing and a spring extending within the biaser housing; a rod; a plate fixed relative to the rod; and an actuator configured to drive the stabilizer toward the deployed position; and wherein the foot is connected to the rod, wherein actuation of the actuator causes the stabilizer to: adjust to the deployed position and apply the predetermined amount of force to the floor, and wherein the spring is configured to act on the biaser housing and the plate to contribute to the predetermined amount of force applied to the floor when the stabilizer is in the deployed position wherein the biaser comprises: a first spring biasing the stabilizer to the retracted position; and a second spring biasing the stabilizer to the deployed position; wherein the biaser comprises: a housing, wherein the first spring extends along an exterior of the housing and the second spring extends along an interior of the housing.
2. The assembly of claim 1, wherein: the surgical cart comprises a surgical cart base; and the biaser comprises: a biaser housing; and a spring configured to act on the surgical cart base and the biaser housing to bias the stabilizer in the retracted position.
3. The assembly of claim 1, further comprising: a force sensor configured to directly or indirectly measure an amount of force applied to the floor when the stabilizer is in the deployed position.
4. The assembly of claim 1, wherein the predetermined amount of force applied to the floor when the stabilizer is in the deployed position is less than a force required to lift the surgical cart off of the floor.
5. The assembly of claim 1, wherein the predetermined amount of force applied to the floor when the stabilizer is in the deployed position is within a range of 80 pounds of force to 100 pounds of force.
6. The assembly of claim 1, further comprising: wherein the stabilizer is a first stabilizer; wherein the assembly further comprises a second stabilizer spaced from the first stabilizer; and wherein the actuator is configured to simultaneously drive the first stabilizer and the second stabilizer toward the deployed position.
7. The assembly of claim 1, further comprising: a mechanical release, wherein the mechanical release is configured to mechanically disengage the actuator and cause the stabilizer to move from the deployed position to the retracted position.
8. The assembly of claim 1, wherein the actuator is configured to drive the stabilizer between the retracted position and the deployed position.
9. A system comprising: a first stabilizer and a second stabilizer for stabilizing a surgical cart on wheels, wherein each of the first and second stabilizers comprise a foot configured to contact a floor to support the surgical cart when the stabilizer is in the deployed position and a biaser configured to bias the stabilizer in the retracted position and contribute to an amount of force the foot applies to the floor in the deployed position; a force applicator extending between the first stabilizer and the second stabilizer; and a linear actuator configured to pull down on the force applicator to cause the first and second stabilizers to move from a retracted position to a deployed

position, wherein the foot of each stabilizer is configured to initially translate with the force applicator and a biaser housing of the biaser when the stabilizer is moving from the retracted position to the deployed position and once the foot of each stabilizer contacts the floor, translate relative to the force applicator and the biaser housing, a force sensor in communication with the linear actuator, the force sensor is configured to sense a force indicative of a force the first stabilizer and the second stabilizer apply to a floor contacting the first stabilizer and the second stabilizer when the first stabilizer and the second stabilizer are in the deployed position; and a controller configured to classify the deployment of the first stabilizer and the second stabilizer as being successful or unsuccessful based on an amount of force detected by the force sensor.

10. The system of claim 9, wherein the force applicator includes: a first end interacting with the first stabilizer; a second end interacting with the second stabilizer; and a portion between the first end and the second end that is coupled to the linear actuator.

11. The system of claim 10, further comprising: wherein the force applicator extends between the linear actuator, the first stabilizer, and the second stabilizer; and wherein the linear actuator is arranged to pull the force applicator toward a floor supporting the surgical cart to cause the first and second stabilizers to move toward the deployed position.

12. The system of claim 9, wherein the linear actuator is configured to adjust between a first predetermined position associated with the retracted position and a second predetermined position associated with the deployed position irrespective of the amount of force the feet of the first and second stabilizers apply to the floor in the deployed position.